Spoken Digit Recognition in Portuguese Using Line Spectral Frequencies

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Abstract. Recognition of isolated spoken digits is the core procedure for a large and important number of applications mainly in telephone based services, such as dialing, airline reservation, bank transaction and price quotation, only using speech. Spoken digit recognition is generally a challenging task since the signals last for short period of time and often some digits are acoustically very similar to each other. The objective of this paper is to investigate the use of machine learning algorithms for digit recognition. We focus on the recognition of digits spoken in Portuguese. However, we note that our techniques are applicable to any language. We believe that the most important task for successfully recognizing spoken digits is the attribute extraction. Audio data is composed by a huge amount of very weak features, and most machine learning algorithms will not be able to build accurate classifiers. We show that Line Spectral Frequencies (LSF) provides a set of highly predictive coefficients for digit recognition. The results are superior than those obtained with state-of-the-art methods using Mel-Frequency Cepstrum Coefficients (MFCC) for digit recognition. In particular, we show that the choice of the right attribute extraction method is more important than the specific classification paradigm, and that the right combination of classifier and attributes can provide almost perfect accuracy.

Keywords: speech recognition, machine learning, spoken digit recognition, line spectral frequency.

1 Introduction

In the last decades, research on speech and speaker recognition has attracted a huge amount of interest, mainly due to the enormous number of applications involving such technology. A few examples are biometric authentication, in which a user voice is used to allow or deny access to a system; and accessibility, in which a user is able to control equipments or navigate on the Internet just using speech, facilitating these tasks to physically-impaired people.

An important speech recognition application, especially useful for telephone service providers, is the recognition of isolated spoken digits. Examples of such services are telephone dialing using speech, airline reservation, banking transactions, price quotation, etc. By using this type of interaction, companies can

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make their services more user-friendly when compared to, for instance, entering numbers on the telephone keypad. This is even more evident when the procedure is done through mobile devices, in which there is not a physically detached keyboard for dialing.

Digit recognition might seem an easy task when compared to recognition of spoken words in general. However, spoken digits recognition is challenging due to two main reasons [9]:

1. Spoken digits are of short acoustic duration, typically a few seconds of speech;
2. Some digits are acoustically very similar to each other (for example, "one" and "nine").

The objective of this paper is to investigate the use of machine learning algorithms for digit recognition. We focus on the recognition of digits spoken in Portuguese. However, we note that our techniques are applicable to any language. We believe that the most important task for successfully recognizing spoken digits is attribute extraction. Machine learning algorithms are fully dependent on predictive attributes to build precise classifiers. In speech recognition, and in sound recognition in general, the raw (audio) data is composed of a huge amount of very weak features, and most machine learning algorithms will not be able to build accurate classifiers, mainly due to the curse of dimensionality.

We show that Line Spectral Frequencies (LSF) provides a set of highly predictive coefficients for digit recognition. The results are superior to those obtained with state-of-the-art methods using Mel-Frequency Cepstrum Coefficients (MFCC) for digit recognition. In particular, we show that the choice of the right attribute extraction method is more important than the specific classification paradigm, and that the right combination of classifier and attributes can provide almost perfect accuracy.

This paper is organized as follows: Section 2 reviews the related work on spoken digit recognition using machine learning techniques. Section 3 briefly introduces the general ideas behind MFCC and LSF. This section also reviews the dynamic windowing pre-processing technique used to extract local coefficients, as well as to generate attribute vectors with fixed length. Section 4 describes the database of spoken digits used in our experiment and the techniques used to segment the data. Section 5 presents our experimental design and discusses the obtained results. Finally, Sect. 6 provides the conclusions and possible future work.

2 Related Work

As previously mentioned, spoken digit recognition is the core of several important applications. Due to its relevance, several papers have been published on this topic for diverse languages such as Japanese [8], English [1], Arabic [2,6], Hindi [13], Bengali [5] and Urdu [3]. In general, these papers have a common framework, in which MFCC are used as principal attributes in conjunction with a classifier induced by a machine learning system.

We scanned the literature searching for papers that proposed approaches and/or evaluated them on spoken digits in Portuguese, and we discovered that there is an enormous lack of research work on that language. To the best of our knowledge, the only pieces of work that considered the Portuguese language were the work of Rodrigues and Trancoso [15], in which MFCC and Hidden Markov Models were used to recognize digits in Portuguese with an accuracy of 99.40%, and, more recently, Bresolin et al. [4], which compared wavelet-based attributes to MFCC for recognizing digits, obtaining an accuracy of 92.97% for the best configuration of attributes based on wavelet and 92.87% for MFCC in speaker-independent digit recognition.

In our work, we show that LSF coefficients provide a more accurate set of attributes than MFCC for digit recognition in Portuguese. With the right combination of LSF coefficients and classifier our approach achieved almost perfect classification accuracy in the same database used in [4].

3 Attribute Extraction Methods

As previously discussed, machine learning algorithms are highly dependent on predictive features in order to build accurate classifiers. In this section we provide a brief description of two well-known methods for attribute extraction, namely Mel-Frequency Cepstrum Coefficients (MFCC) and Line Spectral Frequencies (LSF). The main purpose of these methods is to perform a representational change of the original audio data, from a high-dimensional weak-feature domain to a low-dimensional strong-feature domain.

We start describing the technique of dynamic windowing. The importance of this technique is two-fold: first, the window allows extracting local attributes from smaller parts of the signal and characterizing signal changes in time; second, the dynamic setting allows adjusting the size of the window so that every signal results in the same number of attributes.

3.1 Dynamic Windowing

Usually, speech recognition involves the classification of signals with different durations. This variability occurs not only inter-classes, because different words have different lengths, but also intra-classes, because different speakers usually have different paces. Data with varying length is a problem for several machine learning algorithms that expect a fixed-size attribute-value table as input. We use dynamic windowing as a strategy to overcome that issue.

Dynamic windowing is a simple strategy that breaks a signal of arbitrary length into a set of feature vectors. Each feature vector is the collection of features extracted from a segment of the original signal, which is obtained from a sliding window of width w_s. The value of w_s is dependent on the length of the signal s and the number of windows required. Furthermore, each window has an overlap with the previous, as can be seen in Fig. 1. This overlap must be large enough so that information in the signal transitions are not lost. An overlap higher than 50% is commonly used.

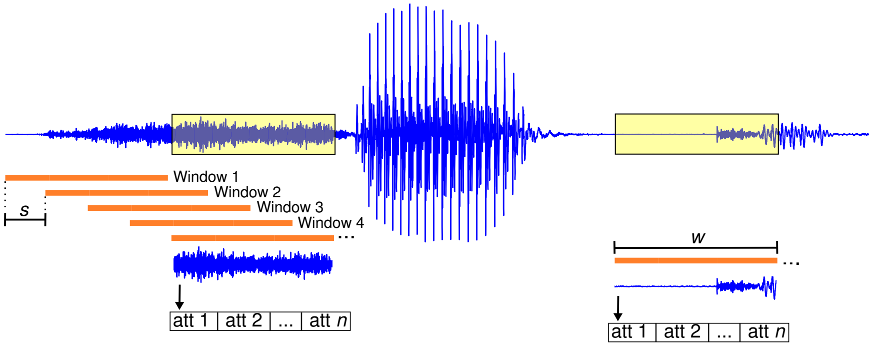

Fig. 1. In the dynamic windowing strategy, the feature extraction uses a sliding window of width w, proportional to a preset number of windows, with an overlap of size s between consecutive windows. So, a $n-$dimentional feature vector is extracted for each window.

In our experiments, we set the window width w_s according to Equation 1 below:

$$w_s = \lceil \frac{e}{1-o} \rceil \tag{1}$$

where o is the overlapping rate in the range between 0 and 1 and e is the window width disregarding overlapping between consecutive windows. The value of e can be obtained from Equation 2 below:

$$e = \lceil \frac{l_s}{n} \rceil \tag{2}$$

where l_s is the signal length and n is the number of windows.

Obviously, the dynamic window strategy should be used with a word of caution: the existence of signals with considerable difference in duration will make the window sizes, and consequently the step sizes, have a large variance. Large step sizes may cause a loss of detail on how the signal evolves in time when features are extracted.

3.2 Mel-Frequency Cepstrum Coefficients

The Mel-Frequency Cepstrum Coefficients are probably the most commonly used attributes in speech processing tasks, such as speaker and speech recognition [18]. Briefly, to calculate those coefficients, we first take the magnitudes of frequency components using an acoustically-defined scale called *mel*, originated from the study of Stevens et al. [16], which relates physical frequencies to the frequencies perceived by the human auditory system. Next, we apply a Discrete Cosine Transform, widely used in data compression [17]. The MFCC are the cepstrum coefficients obtained from this operation. Equation 3 shows the conversion from frequency (f) to mel-frequency (m).

$$m = 2595 \times log_{10}(1 + \frac{f}{700}) \tag{3}$$

3.3 Line Spectral Frequencies

Linear Prediction (LP) is a technique used in many speech applications, such as recognition, compression and modeling [14,10]. LP is based on the fact that a speech signal can be described by Equation 4.

$$\hat{x}_k = \sum_{i=1}^{p} a_i x_{k-i} \tag{4}$$

where k is the time index and p is the order of LP – *i.e.*, the number of employed LP coefficients. The a_i coefficients are calculated in order to minimize the prediction error by means of a covariance or auto-correlation method.

Equation 4 can be rewritten in the frequency domain with a $z-$transform [11]. In this way, a short segment of speech is assumed to be generated as the output of an all-pole filter $H(z) = 1/A(z)$, where $A(z)$ is the inverse filter such that:

$$H(z) = \frac{1}{A(z)} = \frac{1}{1 - \sum_{i=1}^{p} a_i z^{-i}} \tag{5}$$

The Line Spectral Frequencies (LSF) representation, introduced by Itakura [7], is an alternative way to represent LP coefficients. In order to calculate LSF coefficients, the inverse filter polynomial is decomposed into two polynomials $P(z)$ and $Q(z)$:

$$P(z) = A(z) + z^{p+1}A(z^{-1}) \text{ and } Q(z) = A(z) - z^{p+1}A(z^{-1})$$

where $P(z)$ is a symmetric polynomial and $Q(z)$ is an antisymmetric polynomial. The roots of $P(z)$ and $Q(z)$ determine the LSF coefficients.

LSF is well suited for quantization and interpolation [12]. Therefore LSF can represent the speech signal, mapping a large signal to a small number of coefficients, better than other LP representations.

4 Spoken Digits Database

In this section, we describe the database used in our experiments. In order to promote the comparison to literature results and facilitate the reproducibility of our results, we used a publicly available database created by Bresolin et al. [4]. We believe that the use of public data restricts any bias that could possibly arise if we decided to create our own database.

We are also highly committed to reproducibility of our results. Therefore, we created a paper website[1] in which we published every source code and data used in this paper.

[1] `http://www.icmc.usp.br/~diegofsilva/IBERAMIA2012`

4.1 Database Description

The database consists of spoken digits in Portuguese collected during a period of three months, from eighty-two men aged between 18 and 42 years-old. The sampling rate of the recording is $22,050$Hz. Altogether, the database has 216 sequences of 10 digits $(0-9)$ each, totalling 10 classes and $2,160$ examples. Thus, it is a balanced dataset considering the class distribution.

Table 1 presents the approximated pronunciation of the 10 digits in English and by the International Phonetic Alphabet symbols (IPA[2]).

Table 1. Portuguese digits, approximate English pronunciation and IPA symbols

Digit	Portuguese Writing	Pronounce	IPA	Digit	Portuguese Writing	Pronounce	IPA
0	zero	zeh-ro	['zɛru]	5	cinco	seen-coh	['sĩŋku]
1	um	oom	[ũ]	6	seis	say-z	['sejʃ]
2	dois	doy-z	['doiʃ]	7	sete	seh-chee	['sɛtʃi]
3	três	treh-z	['trejʃ]	8	oito	oy-too	['ojtu]
4	quatro	kwah-troh	['kwatru]	9	nove	noh-vee	['nɔvi]

4.2 Signal Pre-processing

Each audio recording is related to one speaker who pronounces all digits from 0 to 9. Therefore, the recordings are not separated for each digit, rather each file

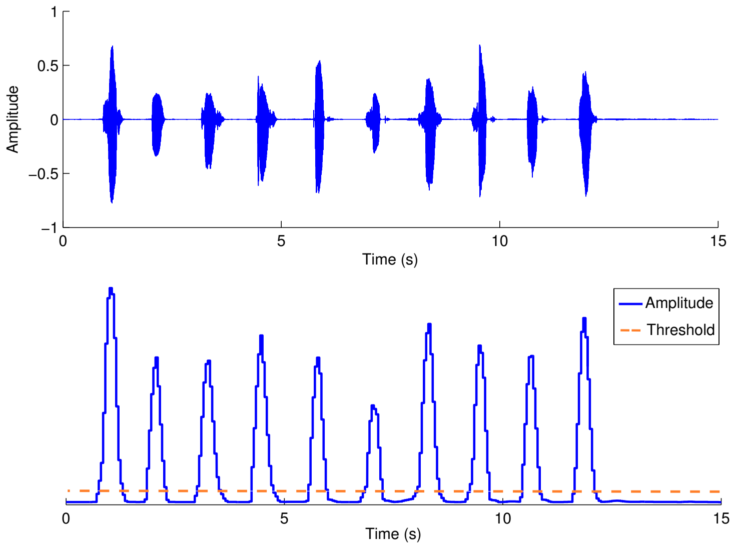

Fig. 2. Segmentation scheme. It's calculated the mean amplitude in the original signal (top) using a sliding window to form a vector that defines a confidence level for the existence of an interesting signal in the window (bottom). Finally, it is set a threshold for acceptance.

[2] http://www.langsci.ucl.ac.uk/ipa/

records one phrase encompassing the pronunciation of all digits. It was necessary to segment each file into 10 fragments, each one containing one digit.

As the signal-to-noise ratio of these signals is relatively high, it was possible to use a simple amplitude-based detector. First, the signal is normalized, dividing the values in each window by the highest value observed. Thus, the relative amplitudes will be within the interval between -1 and 1. After that, the mean amplitude within a sliding window is calculated and used as confidence estimate. The greater the calculated amplitude, the greater the confidence that the window contains part of a spoken signal. Finally, we set an acceptance threshold so that the portions above the threshold are indicative of a spoken digit. We save the segments above the threshold in separate files. This detection method is shown in Fig. 2.

The audio files were generated by speakers pronouncing the digits in numerical order. Thus, the task of labeling signals after segmentation is trivial.

5 Experiments and Results

Our main goal is to show that LSF coefficients can be used as features to provide highly accurate classifiers for spoken digits recognition. We compared MFCC with 13 coefficients against LSF with orders 24 and 48. These numbers of coefficients were used because it is known that 13 MFCC and 12 LP coefficients are sufficient to characterize speech signals. Furthermore, it is common to use a multiple of the number of LP coefficients as the LSF order.

We compare the methods on 12 different scenarios. Each scenario is a different setting of some classification algorithm. The algorithms we use for classification and their related settings are shown in Table 2. We chose these algorithms because they are frequently used and the authors report good classification performance on signal recognition papers. Nearest neighbor classifiers were induced with inverse distance weighting approach.

Table 2. Experiment scenarios description

Scenario	Inducer/settings
1-NN	Nearest Neighbor
5-NN	5-Nearest Neighbor weighted by inverse distance
7-NN	7-Nearest Neighbor weighted by inverse distance
9-NN	9-Nearest Neighbor weighted by inverse distance
SVM-Poly1	Support Vector Machine with Polynomial Kernel with Degree 1
SVM-Poly2	Support Vector Machine with Polynomial Kernel with Degree 2
SVM-Poly3	Support Vector Machine with Polynomial Kernel with Degree 3
SVM-RBF0.01	Support Vector Machine with RBF Kernel wit Gamma=0.01
SVM-RBF0.05	Support Vector Machine with RBF Kernel wit Gamma=0.05
SVM-RBF0.1	Support Vector Machine with RBF Kernel wit Gamma=0.1
NB	Naïve Bayes
RF	Random Forest

For both approaches, MFCC and LSF, features were extracted via dynamic windowing strategy. The width and step size of the sliding window were such that for each signal a set of 25 feature arrays was generated and each adjacent pair of windows had an overlapping area of 75%. Therefore, each feature extraction method generated a dataset where each instance consisted of $25 \times n$ attributes, n being the number of extracted features. In other words, each signal was transformed into an instance with 325, 600 and 1,200 attributes for the 13-MFCC, 24-LSF and 48-LSF strategies, respectively.

Ten-fold cross-validation was used to partition the data sets into training and test sets. In order to reduce results variance by chance we have repeated this process ten times, randomizing the order of examples between two consecutive executions, i.e., we performed 10×10-fold cross-validation.

Our classification results are summarized on Table 3. Values are mean accuracy and standard deviation on accuracy for a set of 10×10-fold cross-validation. Each line stands for an inducer applied on the dataset produced from the features extracted by the method on the column. The most accurate classification algorithm on each method is highlighted.

Table 3. Mean accuracy and standard deviation for the three analyzed methods on the 12 scenarios

Scenario	**MFCC**	**24 LSF**	**48 LSF**
1-NN	86.33 (2.15)	92.92 (1.51)	**93.03 (1.64)**
5-NN	89.52 (1.88)	95.57 (1.19)	**95.66 (1.27)**
7-NN	89.61 (1.85)	95.82 (1.32)	**95.98 (1.37)**
9-NN	90.20 (1.82)	96.13 (1.26)	**95.67 (1.24)**
SVM-Poly1	97.96 (0.86)	98.85 (0.69)	**99.30 (0.57)**
SVM-Poly2	97.88 (0.93)	98.77 (0.70)	**99.31 (0.57)**
SVM-Poly3	97.91 (0.90)	98.75 (0.72)	**99.17 (0.63)**
SVM-RBF0.01	93.62 (1.71)	97.93 (0.95)	**98.64 (0.83)**
SVM-RBF0.05	96.88 (1.17)	98.54 (0.83)	**98.70 (0.77)**
SVM-RBF0.1	97.19 (1.04)	**98.32 (0.88)**	98.02 (0.90)
NB	90.63 (1.66)	**94.86 (1.46)**	94.72 (1.35)
RF	91.83 (1.90)	**96.36 (1.23)**	95.89 (1.37)

Each single execution of the 10-fold cross-validation produced a confusion matrix, from which we were able to assess accuracy values and what are the errors types committed by the classifier. One example of such confusion matrices may be seen on Table 4. This particular matrix shows the results of one of the 10 executions of the SVM-Poly2 inducer on the dataset generated with features extracted by the 48 LSF method. As the original matrix was very sparse, we omitted zeroes from the table for the sake of presentation.

Results show that our extraction methods (24/48 LSF) outperformed the original MFCC extraction method every time. In order to allow for better confidence on the results, we performed the Wilcoxon statistical test. The Wilcoxon test is recommended for comparing pairs of grouped populations. The Wilcoxon test

Table 4. Confusion matrix for one of the test of the SVM-Poly2 inducer on the dataset with features extracted by 48-LSF

		Actual									
		Zero	Um	Dois	Três	Quatro	Cinco	Seis	Sete	Oito	Nove
Predicted	Zero	214							3		
	Um		216							1	
	Dois			214						3	
	Três				216						
	Quatro					215					1
	Cinco						216	1	1		
	Seis							215			
	Sete	2							212		
	Oito			2						212	
	Nove					1					215
Acc. (%)		99.07	100	99.07	100	99.54	100	99.54	98.15	98.15	99.54

indicate that 24 and 48 LSF, despite not being comparable to each other with statistical significance, performed better than the original MFCC approach with 95% of significance for the analyzed dataset and classification algorithms.

6 Conclusions

In this study, we investigated the spoken digit recognition using a database of sounds corresponding to digits spoken in Portuguese. For this task, we investigated the use of LSF coefficients and compared to performance obtained using MFCC, commonly used descriptors in speech recognition. Both descriptors were extracted with dynamic windowing strategy. We used four classification algorithms, with parameter variations.

We demonstrated that the LSF coefficients are highly predictive attributes when classifying spoken digits. Our results showed that these attributes outperformed MFCC for all classifiers analyzed, providing an almost perfect classification performance.

Although the results presented in this study are restricted to the recognition of digits in Portuguese, all the techniques discussed may be used for other speech signals of short duration. Some examples are the recognition of vowels and isolated words.

As future work, we intend to evaluate the use of LSF coefficients in recognition of digits and other related areas in other languages, with the use of dynamic windowing and with Hidden Markov Models and nearest neighbors using the Dynamic Time Warping distance, which does not require that the vectors attributes having the same length.

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